
On a few Aspects of Optical properties of Black holes and Wormholes in General Relativity and Beyond

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DECLARATION

I, Rahul Karmakar bearing registration number 20230000273 of 2023-2024, hereby declare that the project report entitled "**On a few aspects of Optical Properties of Black Hole and Wormholes in General Relativity and Beyond**" being submitted to the Department of Physics, Assam University, Silchar, is based on my own work carried out under the supervision of Prof. Atri Deshamukhya, Department of Physics, Assam University, Silchar. This project report has not been submitted to any other university or institute for the award of any degree.

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CERTIFICATE

This is to certify that the report on the topic **‘On a few Aspects of Optical properties of Black holes and Wormholes in General Relativity and Beyond’** submitted by Mr. Rahul Karmakar, Registration No-20230000273 of 2023-2024 and exam Roll-042523 No-2013900319 is a bonafide piece of work carried out independently under my supervision. This report has been submitted as a requisite component of PHY-654 course and has not been presented elsewhere for any other degree or diploma.

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Abstract

This project carries out a comparative analysis of the optical properties and lensing effects of black holes and wormholes, aiming to identify distinctive optical signatures as observational discriminants. In general relativity and its Kalb-Ramond gravity extension, we analytically derive significant lensing parameters—i.e., deflection angles, photon sphere radii, and shadow profiles for Schwarzschild black holes and Morris-Thorne wormholes. The article begins with an elaborate description of null geodesics in Schwarzschild spacetime and a generalization to black holes in Kalb-Ramond gravity. It measures how deformation parameters influence photon trajectories, shadow radius, and deflection angle.

For wormhole spacetime, we consider the Morris-Thorne geometry and examine lensing for various redshift and shape functions. We find that wormholes can possess multiple photon spheres and produce complex deflection patterns due to the absence of an event horizon, resulting in qualitative differences in image formation from black holes. This report presents a set of theoretically motivated lensing signatures that can be used to guide future observational efforts toward the detection of exotic compact objects. The results demonstrate the strength of gravitational lensing as a diagnostics tool to test modified gravity theories and non-trivial spacetime topologies.

CHAPTER 1

INTRODUCTION

Black holes and wormholes are cosmological structures that occur in deep space, derived from solutions to Einstein's field equations. A black hole is a solution to the Einstein field equation in general relativity which represents a region of space-time with a curvature so extreme that the escape velocity exceeds the speed of light. A black hole consists of an event horizon, a casual boundary beyond which the information cannot propagate to an external observer. Albert Einstein's theory of general relativity was able to theorize that a mass when compressed sufficiently, can deform the space-time and form a black hole. The formation of a black hole can occur due to the natural collapse of a massive star at the end of its life cycle. After formation, it can grow by absorbing mass from its surroundings through the accretion process. Wormholes on the other hand are theoretically proven hypothetical passages connecting two regions of spacetime and have a unique geometry, having two mouths connected by a throat.

Since both black holes and wormholes are gravitationally influenced astronomical objects, therefore they can be mainly studied through their gravitational properties. Gravitational lensing is one of the fundamental phenomena that helps us in researching and identifying the distinct properties of black hole and wormhole spacetimes. This project aims to study the gravitational lensing exhibited by a black hole and a wormhole and a comparative study is carried out to identify the distinct observational signatures that differentiate wormholes from black holes.

1.1 Gravitational lensing

Gravitational lensing is an important phenomenon that can arise due to the curvature of space-time in presence of a massive celestial object such as a star or a galaxy. The massive celestial object bends the path of light as it travels towards the observer, acting like a lens. A lens formed due to the gravitational field can produce maximum deflection of light when it is passing closest to its center and a minimum when it is traveling furthest from its center.

Gravitational lensing can be categorized into three different classes:

Strong Lensing : Distortions are prominently visible in this kind of lensing with the formation of Einstein rings and multiple images.

Weak Lensing : Distortions are less and can be identified with the help of statistical analysis of multiple images in order to detect the coherent distortions of only a few percent.

Microlensing : No visible distortions are detected but there is a change in the background of the light image with the evolution of time.

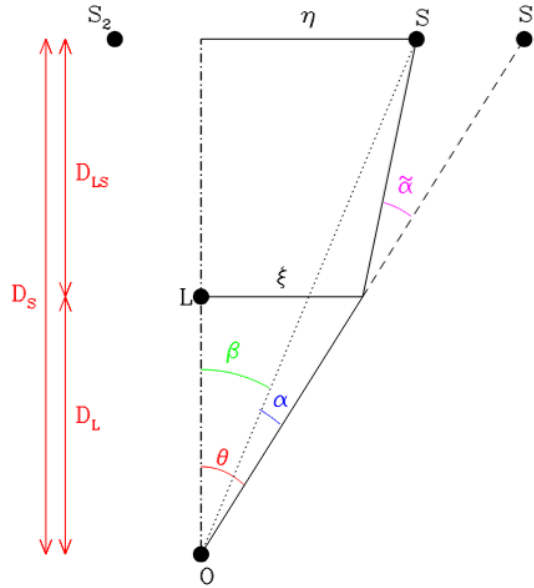


Figure 1.1: Visual representation of gravitational lensing setup (Wambsganss)

The gravitational lens equation (Schneider et al.) is written as :

$$\vec{\beta} = \vec{\theta} - \vec{\alpha}(\theta) = \vec{\theta} - \frac{D_{ds}}{D_s} \vec{\alpha}(D_d(\theta)) \quad (1.1)$$

Where β is the angular position of the source without lensing.

θ is the angular position of the observed image.

D_{ds} is the distance of the lens from the source.

D_s is the distance of the observer from the source.

D_d is the distance of the observer to the lens.

The **deflection angle** in gravitational lensing is the bending of light as it passes near a massive object. The theory of General relativity predicts that a light ray that passes by a spherical body of mass M at a minimum distance ξ is deflected by the "Einstein angle" and is given by :

$$\alpha = \frac{4GM}{c^2\xi} \quad (1.2)$$

Where G is the gravitational constant, M is the mass of the lensing object, c is the speed of light and ξ is the impact parameter. The deflection angle is directly proportional to M , the larger the mass, the greater the deflection, and is inversely proportional to the impact parameter,

For a given point mass, gravitational lensing can produce multiple images or an Einstein ring when the source, lens, and observer are aligned. The Einstein radius is given as :

$$\theta_e = \sqrt{\frac{4GM}{c^2} \frac{D_{LS}}{D_{OL}D_{OS}}} \quad (1.3)$$

The Einstein radius is responsible for defining the angular scale for a lens situation. Reforming the lens equation (1.1) for a single point mass,

$$\vec{\beta} = \theta - \theta_E^2/\theta$$

Solving the above equation for the image position θ tells that an isolated point source will always produce two images of a background source, whose positions can be represented by the two solutions :

$$\theta_{1,2} = \frac{1}{2}(\beta \pm \sqrt{\beta^2 + 4\theta_E^2}) \quad (1.4)$$

The magnification of an image can be defined by the ratio between the solid angles of the image and the source. The solid angle is taken into consideration as the surface brightness remains

conserved. The magnification μ is given as -

$$\mu = \frac{\theta}{\beta} \frac{d\theta}{d\beta} \quad (1.5)$$

For symmetric cases, the image magnification is written as :

$$\mu_{1,2} = \left(1 - \left[\frac{\theta_E}{\theta_{1,2}} \right]^4 \right)^{-1} = \frac{\mu^2 + 2}{2u\sqrt{u^2 + 4}} \pm \frac{1}{2} \quad (1.6)$$

Here, u is defined as the impact parameter which is the angular separation between the lens and the source in units of the Einstein radius $u = \beta/\theta_E$. The total sum of the absolute values of the two image magnification is the measure of total magnification μ and is written as

$$\mu = |\mu_1| + |\mu_2| = \frac{u^2 + 2}{u\sqrt{u^2 + 4}} \quad (1.7)$$

Where the difference between the two image magnifications is unity.

1.2 Black hole spacetime

The Schwarzschild solution was the first exact solution to the Einstein field equation. The solution describes the space-time around a spherically symmetric mass. Birkhoff's theorem states that any spherical symmetric vacuum solution must be static and asymptotically flat and therefore Schwarzschild metric describes the exterior of a non-rotating spherical body.

The Einstein Field Equation (Einstein) relates the curvature of space-time to the distribution of matter and energy and is given as :

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} - \Lambda g_{\mu\nu} = \frac{8\pi g}{c^4}T_{\mu\nu} \quad (1.8)$$

Where $G_{\mu\nu}$ is the Einstein Tensor, defining the curvature of space-time, $R_{\mu\nu}$ is the Ricci curvature tensor, R is the Ricci scalar and Λ is the cosmological constant.

For a vacuum solution (outside the blackhole, where $T_{\mu\nu} = 0$) and assuming $\Lambda = 0$, the equation simplifies to :

$$R_{\mu\nu} = 0$$

The above condition is known as the Vacuum Einstein equation and it implies that the space-time

is Ricci flat.

Considering some assumptions such as spherical symmetry where the space-time is invariant under rotation, meaning the gravitational field depends only on the radial distance r on the center. The space-time is considered to be static, meaning the metric does not change with time and there is no rotation. Static symmetric also implies the existence of a timelike killing vector which is orthogonal spatial hypersurfaces, so that the given metric can be written with no t - r cross terms and all components are time-independent.

The assumptions lead to a matrix in spherical co-ordinates (t, r, θ, ϕ) of the form

$$ds^2 = -A(r)c^2dt^2 + B(r)dr^2 + r^2d\Omega^2 \quad (1.9)$$

where $A(r)$ and $B(r)$ are functions to be determined and $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$. Substituting the metric into $R_{\mu\nu} = 0$ leads to a set of differential equations for $A(r)$ and $B(r)$. Computing the Christoffel symbols $\Gamma_{\alpha\beta}^\mu$ and Ricci Tensor components, we find that

$$B(r) = 1/A(r)$$

where

$$A(r) = 1 - \frac{2GM}{rc^2}$$

Substituting back, we obtain the Schwarzschild metric (Virbhadra and Ellis)

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right)c^2dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (1.10)$$

At $r = r_s = \frac{2GM}{c^2}$, the metric appears to be singular. The radius defines the Schwarzschild radius, the point of no return for infalling matter. At $r = 0$, the curvature becomes infinite, it indicates a physical singularity where general relativity breaks down. Far from the blackhole ($r \gg r_s$), the metric reduces to the Minkowski metric of flat space-time. The Schwarzschild radius itself is simply proportional to mass and every mass M has an event horizon of radius r_s . Outside the horizon ($r > r_s$), the Schwarzschild space-time is static and asymptotically flat. As the metric is static and isotropic, it has no angular momentum, and therefore mass M is the only trivial parameter, making the Schwarzschild metric the simplest blackhole model (Wambsganss).

1.3 Wormhole spacetime

Einstein's formulation of the law of gravitation (Einstein), which is later greatly known as the Einstein Field equation, is based on a certain number of considerations. The field equation is written in the following form:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} - \Lambda g_{\mu\nu} = \frac{8\pi g}{c^4}T_{\mu\nu} \quad (1.11)$$

Where $R_{\mu\nu}$ is the Ricci Curvature tensor and $T_{\mu\nu}$ is the Stress-Energy Tensor and Λ is the Cosmological Constant.

The general form of a line element with spherical symmetry is written in the following form:

$$ds^2 = -e^{\lambda(r,t)}dr^2 - e^{\mu(r,t)}(r^2d\theta^2 + r^2\sin^2\theta d\phi^2) + e^{\nu(r,t)}c^2dt^2 + 2a(r,t)cdrdt \quad (1.12)$$

The above equation can be written in a more simplified form as :

$$ds^2 = -e^{\lambda(r,t)}dr^2 - r^2d\theta^2 - r^2\sin\theta d\phi^2 + e^{\nu(r,t)}c^2dt^2 \quad (1.13)$$

where λ and ν are the functions of r and t .

The components for the Einstein tensor for the metric are computed for $G_t^t, G_r^r, G_\theta^\theta$ and G_ϕ^ϕ . The stress-energy tensor $T_{\mu\nu}$ is taken by considering the diagonal components $T_t^t = \rho, T_r^r = p_r$ and $T_\theta^\theta = T_\phi^\phi = p_t$, where ρ is the energy density, p_r is the radial pressure, and p_t is the lateral pressure.

$$\text{Energy Density, } \rho = \frac{1}{8\pi} \frac{b'(r)}{r^2} \quad (1.14)$$

$$\text{Radial Pressure, } T_r = \frac{1}{8\pi} \left[\frac{b(r)}{r^3} - 2 \frac{\phi'(r) \left(1 - \frac{b(r)}{r}\right)}{r} \right] \quad (1.15)$$

$$\text{Lateral Pressure, } T_\theta = \frac{1}{8\pi} \left(1 - \frac{b(r)}{r}\right) \left[\Phi''(r) + (\Phi'(r))^2 + \frac{\Phi'(r) - \frac{b'(r)}{2r}}{r \left(1 - \frac{b(r)}{r}\right)} \right] \quad (1.16)$$

Substituting these forms into the metric results in the formalism of Morris Thorne metric, given as

$$ds^2 = e^{2\phi(r)}c^2dt^2 - \frac{dr^2}{1 - \frac{b(r)}{r}} - r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (1.17)$$

where, $\phi(r)$ is known as the redshift function, determining the gravitational redshift, $b(r)$ is the

shape function, determining the spatial shape of a wormhole. r is the radial coordinate, and $d\theta^2 + \sin^2 \theta d\phi^2$ are the angular components (Baruah and Deshamukhya).

The wormhole metric is embedded in a higher-dimensional Euclidian space. For the equatorial plane($\theta = \pi/2$), the spatial part of the metric is written as :

$$ds^2 = -e^{2\phi r} dt^2 + \left(1 - \frac{b(r)}{r}\right)^{-1} dr^2 + r^2 d\theta^2$$

Embedding in a cylindrical like co-ordinate system (r, θ, z) ,

$$\frac{dr}{dz} = \pm \left(\frac{r}{b(r)} - 1\right)^{-1/2}$$

The term $\phi(r)$ is responsible for determining the gravitational redshift. For the traversability condition, $\phi(r)$ should remain finite everywhere to avoid event horizon (no infinite redshift).

The term $b(r)$ is known as the shape function and it defines the structural geometry of the wormhole throat. The throat is located at $r = r_0$ where $b(r_0) = r_0$ and r_0 is the minimum radius of the throat. For the flaring-out condition, $b'(r_0) < 1$ where $b(r) = db/dr$. The flaring-out condition ensures that the throat is a minimum radius in an embedding space. For asymptotic flatness ($r \rightarrow \infty, b(r)/r \rightarrow 0$) and the space-time approaches to Minkowski space-time. For further simplification, the redshift function $\phi(r)$ is set to zero so that $e^{2\phi(r)} = 1$ for a zero tidal force scenario. The shape function $b(r)$ is considered as $b(r) = r_0^2/r$, resulting in a simple throat geometry.

Energy conditions are constraints to the stress-energy tensor in general relativity and violation of certain energy conditions is a necessity for satisfying the wormhole solution.

The Null Energy Condition (NEC) states that for any null vector $k^\mu, T_{\mu\nu} k^\mu k^\nu \geq 0$. For Morris Thorne wormhole, the stress-energy components are written as -

$$T_{\mu\nu} k^\mu k^\nu = \rho(r) + \tau(r)$$

The energy density and radial tension of the throat are

$$\rho(r) = \frac{b'(r)}{8\pi r^2}; \tau(r) = -\frac{b(r)}{8\pi r^3}$$

At the throat,

$$\rho + \tau = \frac{b'(r_0)}{8\pi r_0^2} - \frac{1}{8\pi r_0^2} = \frac{b'r_0 - 1}{8\pi r_0^2}$$

For the flare-out condition, $\rho + \tau < 0$. The violation of the Null Energy condition indicates the existence of exotic matter, having negative energy density, and is necessary for the wormhole stability.

1.4 Project objectives

The main objectives for the current project are :

1. To study the optical properties of a black hole in general relativity and Kalb Ramond gravity.
2. To compare the optical properties of the Schwarzschild black hole and Morris Thorne wormhole.

CHAPTER 2

LITERATURE REVIEW

Recent research that are relevant to the work done on this topic are discussed in the section that follows. Among the many works of literature on black holes and wormholes that are now available, we concentrate more on those that are directly relevant to the research that makes up this project. The behavior of black holes under modified gravity and general relativity is the main topic of the first section of the literature review. In-depth research is done on variations in lensing parameters such as photon sphere, shadow radius, and deflection angle. The wormhole metric and its impact on the lensing phenomena are the subject of the second section of the literature review. The primary areas of interest in this study are the photon sphere, shadow radius, and deflection angle.

2.0.1 On black hole gravitational lensing

Gravitational lensing has been a fundamental concept in testing general relativity (GR) since Eddington's observation of light deflection during a solar eclipse. The paper "Gravitational Lensing in Astronomy" (Wambsganss) discusses the early theoretical work on light deflection by gravity which includes Johann Soldner's calculation in 1801 and Albert Einstein's work in 1911 and 1916, confirming Einstein's prediction during the 1919 solar eclipse. The paper derives the lens equation and introduces the concept of Einstein's radius, critical surfaces, mass density, image positions, and magnifications. Key findings include microlensing which reveals compact objects such as stars and planets in galaxies and probes quasar structure. Weak lensing findings are used in describing the cosmic mass distribution.

The basics of gravitational lensing are discussed in the paper "Schwarzschild blackHole Lensing" (Virbhadra and Ellis), highlighting the potential of gravitational lensing in testing theories mainly related to general relativity in strong gravitational fields, particularly around the supermassive black hole at the center of the galaxies. The paper presents the formula for the deflection angle of a light ray in Schwarzschild space-time, calculates the expression for the photon sphere, and provides the expression for the impact parameter. Relativistic images and the formation of Einstein rings are the key findings in this paper and also highlight the difficulties arising while observing due to extinction, scattering, demagnification, and short observation timescales.

Photon sphere and blackhole shadows are extensively studied (Qiao) and a new geometrical approach is introduced to calculate the photon sphere and shadow radius using the optical geometry of the blackhole space-time. The author presents an optical metric for a spherically symmetric black hole. Gaussian curvature and geodesic curvature are defined in two-dimensional surfaces and are used in presenting the formulae for calculating the geodesic curvature in Gaussian curvature. The paper also presents six examples of spherically symmetric black hole solutions including Schwarzschild, Reissner-Nordstrom, and other solutions for which the photon sphere and shadow radius are calculated.

The behavior of photon sphere and blackhole shadow in dynamically evolving space-time are discussed (Mishra et al.) in the framework of general relativity and modified gravity. The author presents the equation of radius for the photon sphere in a static, spherically space-time and derives the differential equation deriving the dynamical evolution of the photon sphere in a general static and spherically symmetric space-time. The photon sphere evolution is mainly studied in Vaidya and Reissner-Nordstrom space-times.

In the paper "Gravitational deflection of light and shadow cast by rotating Kalb-Ramond blackholes" (Kumar et al.), the author describes the Kalb-Ramond field and highlights the non-minimal coupling of Kalb-Ramond field with gravity which leads to Lorentz symmetry violation. The paper investigates the horizon structure and conserved quantities such as linear and angular momentum. The author derives the photon orbit equation in the weak field interaction, calculates

the geodesic curvature of the light curves, and presents an analytical expression for the deflection angle of light in the rotating Kalb-Ramond blackhole space-time.

For the analytical study of gravitational lensing (Chowdhuri et al.), the author highlights the analytical techniques that are used to study gravitational lensing for black holes in general relativity. Equations of motion for light-like geodesics in Schwarzschild spacetimes are calculated and the deflection angle for a light ray is derived. The deflection angle is derived in terms of the minimum radius of the light ray trajectory and mass of the black hole. The author also derives the deflection angle for Kerr-Newmann space-time and presents the observational implication of the strong deflection limit which particularly results in the formation of Einstein rings. The study of the deflection angle helps in studying the multilevel images which helps in predicting how much resolution is required to distinguish between different Einstein rings.

For the Hairy Schwarzschild blackhole (Jha and Rahaman), gravitational lensing is used to study the geometry of the event horizon and properties of the black hole, including the mass, charge, and angular momentum. The concept of gravitational decoupling is introduced which allows the study of deformation of known solutions in general relativity. The Hairy metric is presented and the horizon radius is calculated which decreases with an increase in α (deformation factor). The deflection angle is calculated for the strong field limit and a variation of the deflection angle is presented which decreases with an increase in the deformation parameter.

Another paper addressing the gravitational lensing of a Schwarzschild-like blackhole in Kalb-Ramond gravity (Junior et al.) derives the exact analytical expression of deflection angle but in terms of elliptic integrals. The derivation starts with a generally spherically symmetric line element and introduces two conserved quantities- linear and angular momentum and the deflection angle is obtained by calculating the impact parameter b in terms of the distance of the closest approach. The lens equation is introduced in the weak field limit, where the source and observer are far from the lens, and light rays are slightly distorted. Strong deflection angles are derived in scenarios where light rays pass very close to the photon sphere of the black hole. The logarithmic expansion of the deflection angle helps in analyzing the geometrical properties of the motion of light rays.

For the strong gravitational lensing in the Kalb-Ramond field (Chakraborty and SenGupta), the deflection angle suffered by a photon is derived as it travels from asymptotic infinity to the impact distance. The strong gravitational lensing is set up in the generally spherically symmetric space-time. The observables associated with gravitational lensing include the deflection angle of the innermost image, the angular separation between the outermost and innermost images, and the astronomical magnitude difference between the outermost and the innermost images. The Kalb-Ramond field is present in four-dimensional space-time and additional spatial dimensions are introduced in order to study the lensing signatures in the effective four-dimensional space.

2.0.2 On wormhole gravitational lensing

While most of the gravitational lensing research is based on conventional astrophysical objects, theoretical advancements were also made in studying the lensing effects caused by exotic space-time geometries, particularly wormholes. Albert Einstein and Nathan Rosen were the first to detect the existence of such structures. In the paper "The Particle Problem in General Theory of Relativity" (Einstein and Rosen), a model of matter and electricity was studied which was purely based on fields and is described by the general relativity and Maxwell theory. Attempts were made to eliminate the existing singularities or to obtain a singularity-free solution in static, spherically symmetric cases. Modifications were introduced in the metric in order to make sure that the equation remains regular and finite. The elimination of the singularities from the Schwarzschild metric transforms the metric and treats the space-time as two congruent layers, connected by a common space, later known as the Einstein-Rosen bridge.

Einstein-Rosen bridges contained a fundamental problem - lack of traversability. In attempts to make wormholes exchange matter and information, certain modifications were introduced to the spherically symmetric, static metric, and fundamental properties were defined. The paper "Wormholes in space-time and their use for Interstellar Travel- a tool for teaching General Relativity" (Morris and Thorne) suggested how Schwarzschild wormholes are non-traversable and are not fit for interstellar travel due to high tidal force at the throat of the wormhole. The desired properties for a traversable wormhole are such that the metric should obey the Einstein field equations and the throat should not contain any horizon. For the stability of the throat, the concept of exotic matter was introduced having negative energy density and negative pressure density, violating the null energy condition and countering the high tidal forces of the throat.

The concept of traversable wormholes is further discussed in the paper "Natural Wormholes as gravitational lenses" (Cramer et al.) where wormholes are framed by exotic cosmic strings with negative mass density. Inflation mechanisms in the early universe were accounted as one of the possible scenarios for the formation of wormholes. The paper also studies the deflection of light due to positive and negative mass. Objects having negative mass were found to produce unique lensing effects at certain configurations and show the possibility of natural wormholes, having net negative mass.

The phenomenon of gravitational lensing caused by traversable Lorentzian wormholes is discussed in the paper "Gravitational Lensing by Wormholes" (Nandi et al.). The authors study gravitational lensing by considering two theoretical frameworks: The Einstein minimally coupled scalar field (EMS) and the Brane world model. The gravitational lensing effects are analyzed in the strong field regime and are used in detecting singularities. EMS theory is used in detecting lensing caused by massive and massless wormholes in minimally coupled scalar fields and the Brane world model is used to study lensing in higher dimensional bulk space-time. Strong field approximation is used to derive the deflection angle by light rays and is found that the deflection angle increases with an increase in wormhole scalar field which results in more significant light bending.

Morris Thorne wormholes are further studied (Shaikh et al., "A novel gravitational lensing feature by wormholes") in order to detect unique gravitational lensing features. The study suggests that Morris Thorne wormholes are found to exhibit two-photon spheres where one is found outside the wormhole throat and the other one is found at the throat itself. The two-photon spheres result in the formation of two distinct sets of Einstein rings where rings associated with the outer photon sphere are less tightly packed and rings associated with the throat photon sphere are more densely packed.

The same authors in another paper "Strong gravitational lensing by wormholes" explore the scenarios where the observer and light source are either on the same side of the throat or on the opposite side of the throat. Bozza's strong field approximation is used to compute the deflection angle and results suggest that a wormhole can have a photon and an anti-photon sphere on both

sides of the throat which can lead to novel lensing configurations. Certain scenarios suggest that the throat itself can act as a photon sphere. The symmetry of the images formed by the photon sphere can provide a unique lensing signature for wormholes.

Shadow and weak gravitational lensing are studied for Ellis-Bronnikov wormholes where photon trajectories are examined using Hamilton-Jacobi formalism in the presence of plasma (Alloqulov et al.). Shadow analysis is done by calculating the shadow radius R_{sh} using the angular radius of the photon sphere r_{ph} . Results show that the shadow size increases with the wormhole parameter α and decreases with an increase in plasma frequency. The deflection angle tends to decrease with an increase in impact parameter b and the total magnification of the lensed image increases with the plasma parameter values, suggesting that uniform plasma enhances the deflection angle and image magnification. In the paper "Strong Deflection Limit Analysis and Gravitational Lensing of an Ellis Wormhole" (Tsukamoto), the authors suggest that the deflection angle in an Ellis wormhole diverges logarithmically near the critical impact parameter which results in the production of multiple images with the innermost relativistic image having the ability to determine the wormhole throat size.

CHAPTER 3

SCHWARZSCHILD BLACK HOLE LENSING

The **Schwarzschild metric** is given by the following expression :

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (3.1)$$

For equatorial motion, we will consider $\theta = \pi/2$. The above metric simplifies to the following form

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\phi^2 \quad (3.2)$$

We define a set of killing vectors ξ^μ which will be used to represent the symmetries of space-time. For a metric containing a killing vector, the momentum remains conserved. Since the Schwarzschild metric is time-independent, therefore the time component g_{tt} remains conserved. For a particle, the Lagrangian $\mathcal{L}$ is given as

$$L = - \left(1 - \frac{2GM}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2$$

For the time component of generalized momentum,

$$\begin{aligned} p_t &= \frac{\partial \mathcal{L}}{\partial \dot{t}} = g_{tt} \dot{t} \\ g_{tt} &= \left(1 - \frac{2GM}{r}\right) \dot{t} \\ \Rightarrow P_t &= \left(1 - \frac{2GM}{r}\right) \frac{dt}{d\lambda} \end{aligned}$$

Since the ξ^μ remains conserved, therefore $p_t = -E$

$$E = \left(1 - \frac{2GM}{r}\right) \frac{dt}{d\lambda} \quad (3.3)$$

where λ is known as the affine parameter that preserves the natural structure of the space. Now for the ϕ component of Generalized momentum, we have

$$\begin{aligned} p_\phi &= \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = g_{\phi\phi} \dot{\phi} = r^2 \dot{\phi} \\ \Rightarrow r^2 \dot{\phi} &= L \end{aligned}$$

For a light-like vector, we consider $ds^2 = 0$. The geodesic equation for the Schwarzschild metric becomes

$$0 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\phi^2 \quad (3.4)$$

Substituting the values from conserved energy and conserved angular momentum and considering the impact parameter $b = L/E$, we get

$$\begin{aligned} b &= \frac{r^2 d\phi/dt}{\left(1 - \frac{2GM}{r}\right) dt/d\lambda} \\ \frac{b}{r^2} \left(1 - \frac{2GM}{r}\right) &= \frac{d\phi}{dt} \end{aligned} \quad (3.5)$$

Substituting $d\phi/dt$ in geodesic equation, we get

$$\begin{aligned} 0 &= - \left(1 - \frac{2GM}{r}\right) + \left(1 - \frac{2GM}{r}\right)^{-1} \left(\frac{dr}{dt}\right)^2 + r^2 \left(\frac{b}{r^2} \left(1 - \frac{2GM}{r}\right)\right)^2 \\ \Rightarrow \left(\frac{dr}{dt}\right)^2 &= \left(1 - \frac{2GM}{r}\right)^2 - \frac{b^2}{r^2} \left(1 - \frac{2GM}{r}\right)^3 \\ \Rightarrow 0 &= E^2 + \left(1 - \frac{2GM}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} \end{aligned} \quad (3.6)$$

For light deflection, we consider $u = 1/r$

$$\begin{aligned} \frac{dr}{d\lambda} &= \frac{dr}{du} \frac{du}{d\lambda} = -\frac{1}{u^2} \frac{du}{d\lambda} \\ 0 &= -E^2 + (1 - 2GMu)^{-1} \left(-\frac{1}{u^2} \frac{du}{d\lambda}\right)^2 + L^2 u^2 \end{aligned}$$

$$\begin{aligned}
&\Rightarrow 0 = -E^2 + (1 - 2GMu)^{-1} \left(\frac{1}{u^4} \left(\frac{du}{d\lambda} \right)^2 \right) + L^2 u^2 \\
&\Rightarrow 0 = -E^2 u^4 (1 - 2GMu) + \left(\frac{du}{d\lambda} \right)^2 + L^2 u^6 (1 - 2GMu) \\
&\quad \therefore \frac{du}{d\lambda} = \frac{du}{d\phi} \frac{d\phi}{d\lambda} = Lu^2 \\
&\Rightarrow 0 = -E^2 u^4 (1 - 2GMu) + \left(\frac{du}{d\phi} Lu^2 \right)^2 + L^2 u^6 (1 - 2GMu) \\
&\Rightarrow 0 = -\frac{E^2}{L^2} (1 - 2GMu) + \left(\frac{du}{d\phi} \right)^2 + L^2 u^6 (1 - 2GMu) \\
&\Rightarrow 0 = -\frac{1}{b^2} (1 - 2GMu) + \left(\frac{du}{d\phi} \right)^2 + u^2 (1 - 2GMu) \\
&\quad \left(\frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2 + 2GMu
\end{aligned} \tag{3.7}$$

Equation (3.7) describes the orbit equation for light in the Schwarzschild metric and is written in terms of $u = 1/r$ and angular coordinate ϕ .

Now we consider the weak field approximation ($2GMu \ll 1$) For the zeroth order solution which describes the case for no deflection, the $2GMu$ is considered as zero. Then equation (3.7) becomes (considering the natural system of units)

$$\left(\frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2 \tag{3.8}$$

The above equation resembles the equation for a straight line in polar coordinates. Solving the equation

$$\begin{aligned}
&\left(\frac{du}{d\phi} \right) = \pm \sqrt{\left(\frac{1}{b^2} - u^2 \right)} \\
&\Rightarrow \int \frac{du}{d\phi} = \int \pm \sqrt{\left(\frac{1}{b^2} - u^2 \right)} \\
&\Rightarrow \int \frac{du}{\sqrt{\left(\frac{1}{b^2} - u^2 \right)}} = \pm \int d\phi \\
&\therefore \int \frac{du}{\sqrt{\left(\frac{1}{b^2} - u^2 \right)}} = \arcsin(bu) + c \\
&\Rightarrow \arcsin(bu) = \pm \phi + c
\end{aligned}$$

$$\Rightarrow bu = \sin(\pm\phi + c)$$

$$\Rightarrow u = \frac{1}{b} \sin(\phi + c)$$

For simplicity, we consider the constant $c = 0$

$$u_0(\phi) = \frac{1}{b} \sin(\phi) \quad (3.9)$$

The above equation is the solution of the differential equation. For the closest approach, $\phi = \frac{\pi}{2}$ and therefore $u_0(\pi/2) = 1/b$

For the first-order correction, we assume the solution of the form

$$u(\phi) = u_0(\phi) + u_1(\phi) \quad (3.10)$$

where $u_1(\phi)$ is the first order correction due to perturbation. Equation (3.7) is then written as

$$\begin{aligned} \left(\frac{du_0}{d\phi} + \frac{du_1}{d\phi} \right)^2 &= \frac{1}{b^2} - (u_0 + u_1)^2 + \frac{2GM}{b^3} \sin^3 \phi \\ \Rightarrow \left(\frac{du_0}{d\phi} \right)^2 + \left(\frac{du_1}{d\phi} \right)^2 + 2 \frac{du_0}{d\phi} \frac{du_1}{d\phi} &= \frac{1}{b^2} - (u_0)^2 - (u_1)^2 - 2u_0 u_1 + \frac{2GM}{b^3} \sin^3 \phi \\ \therefore \left(\frac{du_0}{d\phi} \right)^2 &= \frac{1}{b^2} - u_0^2 \\ \frac{du_0}{d\phi} \frac{du_1}{d\phi} &= -u_0 u_1 + \frac{GM}{b^3} \sin^3 \phi \end{aligned}$$

Using $u_0(\phi) = (1/b) \sin \phi$ and $\frac{du_0}{d\phi} = (1/b) \cos \phi$, we get

$$\begin{aligned} \Rightarrow \frac{1}{b} \cos \phi \frac{du_1}{d\phi} &= -\frac{1}{b} \sin \phi u_1 + \frac{GM}{b^3} \sin^3 \phi \\ \Rightarrow \cos \phi \frac{du_1}{d\phi} &= -\sin \phi u_1 + \frac{GM}{b^2} \sin^3 \phi \end{aligned} \quad (3.11)$$

Equation (3.11) is the first order linear differential equation and its solution is given as

$$u_1(\phi) = \frac{GM}{b^2} (1 + \cos^2 \phi) \quad (3.12)$$

The above equation represents the first-order corrected solution. The total deflection angle for the weak field approximation is written as

$$\alpha \approx 2 \left(\frac{2GM}{b} \right) = \frac{4GM}{b} \quad (3.13)$$

Which resembles the deflection angle of light in the weak field limit of a Schwarzschild black hole.

CHAPTER 4

KALB-RAMOND GRAVITY - AN EXTENSION TO GENERAL RELATIVITY

The Kalb Ramond field can be defined as an antisymmetric second-rank tensor field $B_{\mu\nu} = -B_{\nu\mu}$ that arises naturally in the string theory. It is the generalization of the electric gauge field A_μ with gauge transformation

$$B_{\mu\nu} = \partial_{[\mu}\Lambda_{\nu]}$$

where Λ_μ is the vector gauge potential.

Under string theory, the field is interpreted as a higher gauge field coupling the string world sheet, which is a generalization of the electromagnetic potential for the point particles. The Kalb Ramond field strength is of the three forms:

$$H_{\mu\nu\rho} = \partial_{[\mu}\beta_{\nu\rho]}$$

which is invariant under gauge transformation.

In four dimensions, the field strength $H_{\mu\nu\rho}$ can be dualized to a pseudo-scalar ("axion") field. The Kalb Ramond field can be naturally treated as a rank 2 field. The low energy action for gravity and the Kalb Ramond field typically contains the Einstein Hilbert term and a kinetic term for H. The common form of action can be given as:

$$S = d^4x \sqrt{-g} \left[\frac{1}{2}R - \frac{1}{12}H_{\mu\nu\rho}H^{\mu\nu\rho} + \dots \right] + S_{matter} \quad (4.1)$$

where R is the Ricci scalar and $-\frac{1}{12}H^2$ is the Kinetic term for the antisymmetric field (Duan et al.).

$$S = \frac{1}{2} \int d^4x \sqrt{-g} R - \frac{1}{12} \int d^4x \sqrt{-g} H_{\mu\nu\rho} H^{\mu\nu\rho} + S_{matter}$$

Changing this action yields the Einstein field equation :

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = T_{\mu\nu}^M + T_{\mu\nu}^{KR} \quad (4.2)$$

where $T_{\mu\nu}^{KR}$ is the stress-energy tensor for the Kalb Ramond field.

$$T_{\mu\nu}^{KR} = \frac{1}{2} H_{\mu\alpha\beta} H_{\nu}^{\alpha\beta} - \frac{1}{12} g_{\mu\nu} H^{\alpha\beta\gamma} H_{\alpha\beta\gamma} + \dots$$

plus any contribution from the potential or non-minimal coupling.

The Einstein-Cartan theory allows space-time torsion $T_{\mu\nu}^\rho$ which is an antisymmetric part of the affine connection. A Kalb Ramond field naturally introduces such torsion (Paul and SenGupta) and can be identified as

$$T_{\mu\nu}^\rho \approx H_{\mu\nu}^\rho \quad (4.3)$$

The Einstein-Hilbert action can be extended in Kalb Ramond terms and possible couplings. The general form includes the Einstein Hilbert term $\frac{1}{2}R$, cosmological constant Λ , Kalb Ramond kinetic term $-\frac{1}{12}H_{\mu\nu\rho}H^{\mu\nu\rho}$ and a potential $V(B^2)$ that fixes a non-zero vacuum value. The four-dimensional action can be written as :

$$S = \frac{1}{2} \int d^4x \sqrt{-g} \left[R - 2\Lambda - \frac{1}{6} H^{\mu\nu\rho} H_{\mu\nu\rho} - V(B^2) + \xi_2 B^{\rho\mu} B_\mu^\nu R_{\rho\nu} + \xi_3 B^{\mu\nu} B_{\mu\nu} R \right] + S \quad (4.4)$$

where $\xi_2 B^{\rho\mu} B_\mu^\nu R_{\rho\nu}$ and $\xi_3 B^{\mu\nu} B_{\mu\nu} R$ are the non-minimal curvature couplings. The modified Einstein equation is written as -

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = T_{\mu\nu}^{matter} + T_{\mu\nu}^{KR} \quad (4.5)$$

where $T_{\mu\nu}^{KR}$ contains terms for H^2 .

The Kalb Ramond field enters as a new bosonic field, having its stress energy and can contribute to torsion and Lorentz violating effects in the gravitational dynamics (Baruah et al.).

4.1 Black hole lensing in Kalb-Ramond gravity

The Hairy (Pseudo-Electric) metric is a solution to the black holes related to the no-hair theorem, meaning black holes can be described in only three parameters: mass, charge, and angular momentum. In modified gravity, blackholes can exhibit additional structures known as "hair" which can be described in scalar or vector fields and the solutions can be treated as **hairy blackholes**.

The **Schwarzschild** metric is given as

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (4.6)$$

Under the influence of **Kalb-Ramond Gravity**, the equation modifies to :

$$ds^2 = - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right) dt^2 + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right)^{-1} dr^2 + d\Omega^2$$

$$ds^2 = - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (4.7)$$

Considering the equatorial plane, $\theta = \pi/2$.

$$ds^2 = - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right) dt^2 + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right)^{-1} dr^2 + r^2 d\phi^2$$

The Lagrangian $\mathcal{L}$ is given as :

$$\mathcal{L} = - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2 \quad (4.8)$$

Deriving the conserved energy and conserved angular momentum, we get

$$P_t = g_{tt} \frac{dt}{d\lambda}$$

$$\Rightarrow g_{tt} = - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right)$$

$$\Rightarrow P_t = -E$$

$$\Rightarrow E = \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}}\right) \frac{dt}{d\lambda} \quad (4.9)$$

$$\begin{aligned}
P_\phi &= g_{\phi\phi} \frac{d\phi}{d\lambda} \\
\Rightarrow g_{\phi\phi} &= r^2; \quad P_\phi = L \\
\Rightarrow L &= r^2 \frac{d\phi}{d\lambda}
\end{aligned} \tag{4.10}$$

Substituting the values of $dt/d\lambda$ and $\frac{d\phi}{d\lambda}$ in the above equation, we get :

$$\begin{aligned}
0 &= - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right) \frac{E^2}{\left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right)} + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right)^{-1} \left(\frac{dr}{d\lambda} \right)^2 + r^2 \left(\frac{L^2}{r^4} \right) \\
\Rightarrow & - \frac{E^2}{\left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right)} + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right)^{-1} \left(\frac{dr}{d\lambda} \right)^2 + \frac{L^2}{r^2} = 0 \\
\Rightarrow & -E^2 + \left(\frac{dr}{d\lambda} \right)^2 + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right) \frac{L^2}{r^2} = 0 \\
\Rightarrow & \left(\frac{dr}{d\lambda} \right)^2 = E^2 - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right) \frac{L^2}{r^2}
\end{aligned} \tag{4.11}$$

Now we define an impact parameter $b = L/E$.

$$\begin{aligned}
\left(\frac{dr}{d\lambda} \right)^2 &= E^2 - \frac{E^2 b^2}{r^2} \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{2/\lambda}} \right) \\
\Rightarrow \frac{1}{E^2} \left(\frac{dr}{d\lambda} \right)^2 &= 1 - \frac{b^2}{r^2} \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{2/\lambda}} \right)
\end{aligned}$$

Let $u = 1/r$, then

$$\begin{aligned}
\frac{dr}{d\lambda} &= \frac{dr}{du} \frac{du}{d\lambda} = -\frac{1}{u^2} \frac{du}{d\lambda} \\
\Rightarrow \frac{1}{E^2} \left(-\frac{1}{u^2} \frac{du}{d\lambda} \right)^2 &= 1 - b^2 u^2 (1 - 2GMu + \Gamma u^{2/\lambda}) \\
\Rightarrow \frac{1}{E^2 u^4} \left(\frac{du}{d\lambda} \right)^2 &= 1 - b^2 u^2 (1 - 2GMu + \Gamma u^{2/\lambda})^2
\end{aligned}$$

We know, $(d\phi/d\lambda) = L/r^2 = Lu^2$

$$\begin{aligned}
\frac{du}{d\lambda} &= \frac{du}{d\phi} \frac{d\phi}{d\lambda} = \frac{du}{d\phi} Lu^2 \\
\Rightarrow \frac{1}{E^2 u^4} \left(\frac{du}{d\phi} Lu^2 \right)^2 &= 1 - b^2 u^2 (1 - 2GMu + \Gamma u^{2/\lambda})^2
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \frac{L^2}{E^2} \left(\frac{du}{d\phi} \right)^2 L^2 u^4 = 1 - b^2 u^2 (1 - 2GMu + \Gamma u^{2/\Lambda})^2 \\
&\Rightarrow b^2 \left(\frac{du}{d\phi} \right)^2 = 1 - b^2 u^2 (1 - 2GMu + \Gamma u^{2/\Lambda})^2 \\
&\Rightarrow \left(\frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2 (1 - 2GMu + \Gamma u^{2/\Lambda})^2
\end{aligned} \tag{4.12}$$

The above equation represents the orbit equation in the modified metric. For the weak field approximation, we consider $2GMU \ll 1$. For the zeroth order solution, we assume $2GMu = 0$ and $\Gamma = 0$. The equation then modifies into

$$\left(\frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2$$

Solving the equation:

$$\begin{aligned}
&\frac{du}{d\phi} = \pm \sqrt{\frac{1}{b^2} - u^2} \\
&\Rightarrow \int \frac{du}{\sqrt{\frac{1}{b^2} - u^2}} = \pm \int d\phi \\
&\because \int \frac{du}{\sqrt{\frac{1}{b^2} - u^2}} = \arcsin(bu) + c \\
&\Rightarrow \arcsin(bu) = \pm \phi + c \\
&\Rightarrow bu = \sin(\pm \phi + c) \\
&\Rightarrow u = \frac{1}{b} \sin(\phi + c) \\
&u_0(\phi) = \frac{1}{b} \sin(\phi)
\end{aligned} \tag{4.13}$$

The above equation is the solution of the differential equation. For the closest approach, $\phi = \frac{\pi}{2}$, and therefore $u_0(\pi/2) = \frac{1}{b}$.

For the first order correction, we have

$$\left(\frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2 + 2GMu^3 - \Gamma u^{(2+2/\Lambda)}$$

Let the solution be of the form $u(\phi) = u_0(\phi) + u_1(\phi)$

$$\left(\frac{du_0}{d\phi} + \frac{du_1}{d\phi}\right)^2 = \frac{1}{b^2} - (u_0 + u_1)^2 + 2GM(u_0 + u_1)^3 + \Gamma(u_0 + u_1)^{2+2/\lambda}$$

Considering the following assumptions :

$$\left(\frac{du_0}{d\phi} + \frac{du_1}{d\phi}\right)^2 \approx \left(\frac{du_0}{d\phi}\right)^2 + 2\frac{du_0}{d\phi}\frac{du_1}{d\phi}$$

$$(u_0 + u_1)^2 \approx u_0^2 + 2u_0u_1$$

$$(u_0 + u_1)^3 \approx u_0^3 + 3u_0^2u_1$$

$$(u_0 + u_1)^{2+2/\lambda} \approx u_0^{2+2/\lambda} + (2 + 2/\lambda)u_0^{1+2/\lambda}u_1$$

$$\Rightarrow \left(\frac{du_0}{d\phi}\right)^2 + 2\frac{du_0}{d\phi}\frac{du_1}{d\phi} = \frac{1}{b^2} - u_0^2 - 2u_0u_1 + 2GM(u_0^3 + 3u_0^2u_1) - \Gamma(u_0^{2+2/\lambda} + \left(2 + \frac{2}{\lambda}\right)u_0^{1+2/\lambda}u_1)$$

We know,

$$\left(\frac{du}{d\phi}\right)^2 = \frac{1}{b^2} - u^2$$

$$\Rightarrow \frac{1}{b^2} - u_0^2 + 2\frac{du_0}{d\phi}\frac{du_1}{d\phi} = \frac{1}{b^2} - u_0^2 - 2u_0u_1 + 2GM(u_0^3 + 3u_0^2u_1) - \Gamma(u_0^{2+2/\lambda} + \left(2 + \frac{2}{\lambda}\right)u_0^{1+2/\lambda}u_1)$$

$$\Rightarrow 2\frac{du_0}{d\phi}\frac{du_1}{d\phi} = -2u_0u_1 + 2GM(u_0^3 + 3u_0^2u_1) - \Gamma(u_0^{2+2/\lambda} + (2 + 2/\lambda)u_0^{1+2/\lambda}u_1)$$

We know $u_0\phi = \frac{1}{b}\sin\phi$; $\frac{du_0}{d\phi} = \frac{1}{b}\cos\phi$

$$\Rightarrow 2\left(\frac{1}{b}\cos\phi\right)\frac{du_1}{d\phi} = -2\left(\frac{1}{b}\sin\phi\right)u_1 + 2GM\left(\left(\frac{1}{b}\sin\phi\right)^3 + 3\left(\frac{1}{b}\sin\phi\right)^2u_1\right) - \Gamma\left(\left(\frac{1}{b}\sin\phi\right)^{2+2/\lambda} + (2 + 2/\lambda)\left(\frac{1}{b}\sin\phi\right)^{1+2/\lambda}u_1\right)$$

$$\Rightarrow \frac{du_1}{d\phi} = -\frac{\sin\phi}{\cos\phi}u_1 + \frac{GM}{b^2}\frac{\sin^3\phi}{\cos\phi} + \frac{3GM}{b}\frac{\sin^2\phi}{\cos\phi}u_1 - \frac{\Gamma}{2b^{1+2/\lambda}\frac{\sin^{2+2/\lambda}}{\cos\phi}\phi} - \frac{\Gamma}{2b^{2/\lambda}}(2+2/\lambda)\frac{\sin^{1+2/\lambda}}{\cos\phi}u_1$$

$$\Rightarrow \frac{du_1}{d\phi} = -\tan\phi u_1 + \frac{GM}{b^2}\sin^2\phi\tan\phi + \frac{3GM}{b}\sin\phi\tan\phi u_1 - \frac{\Gamma}{2b^{1+2/\lambda}}(2 + 2/\lambda)\sin^{2/\lambda}\phi\tan\phi$$

The above equation is a differential equation. Differentiating the orbit equation with respect to ϕ . In order to obtain the deflection angle, we get

$$\begin{aligned}
2 \frac{du}{d\phi} \frac{d^2u}{d\phi^2} &= \frac{d}{d\phi} \left[\frac{1}{b^2} - u^2 + 2GMu^3 - \Gamma u^{2+2/\lambda} \right] \\
\Rightarrow 2 \frac{du}{d\phi} \frac{d^2u}{d\phi^2} &= -2u \frac{du}{d\phi} + 6GMu^2 \frac{du}{d\phi} - \Gamma(2 + 2/\lambda)u^{(2/\lambda+1)} \\
\Rightarrow \frac{d^2u}{d\phi^2} + u &= \frac{GM}{b^2} + \frac{\Gamma}{2b^{1-2/\lambda}} \sin^{2/\lambda-1}
\end{aligned}$$

Now, $u = u_0 + \delta u = \frac{\sin\phi}{b} + \delta u$

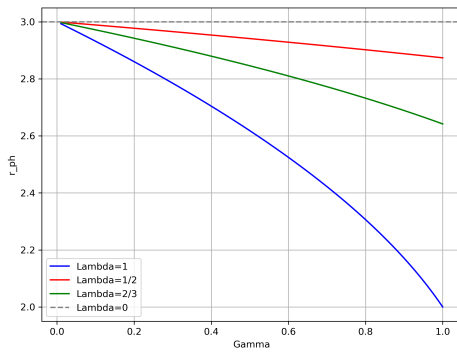
$$\begin{aligned}
\Rightarrow \frac{d^2(\delta u)}{d\phi^2} + \delta u &= \frac{GM}{b^2} + \frac{\Gamma}{2b^{1-2/\lambda}} \sin^{2/\lambda-1} \phi \\
\Rightarrow \alpha &= \frac{4GM}{b} - \frac{\pi\Gamma}{2b^{1-2/\lambda}}
\end{aligned} \tag{4.14}$$

Which is the required equation for the deflection angle.

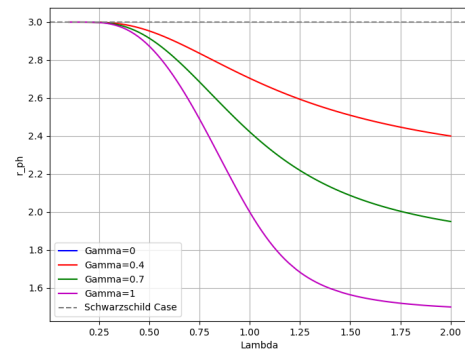
The above equation follows the Einstein deflection angle and describes the deflection of light in the weak field approximation.

CHAPTER 5

PHOTON SPHERE, SHADOW RADIUS AND WEAK DEFLECTION ANGLE IN GENERAL RELATIVITY AND KALB-RAMOND GRAVITY



(a) Variation in Gamma



(b) Variation in Lambda

Figure 5.1: Photon sphere variation for (a)Fixed lambda values (b)Fixed Gamma values

The **photon sphere** is a spherical region around a blackhole where photons or light rays travel in unstable circular orbits due to extreme space-time curvature.

For the line element, we consider the equatorial plane, i.e. $\theta = \pi/2$. There exists a null geodesic for that given space-time which is circular and that will essentially act like a photon sphere. Since the trajectory of null geodesics is circular, we set $r = \text{constant} \equiv r_{ph}$. Being null geodesic, we know $ds^2 = 0$ and our line element modifies to

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2 + r^2 d\phi^2 \quad (5.1)$$

$$ds^2 = - \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right) dt^2 + \left(1 - \frac{2GM}{r} + \frac{\Gamma}{r^{\frac{2}{\lambda}}} \right)^{-1} dr^2 + r^2 d\phi^2 \quad (5.2)$$

The photon sphere radius can be derived from a given line element by using the following relation (Gan et al.) :

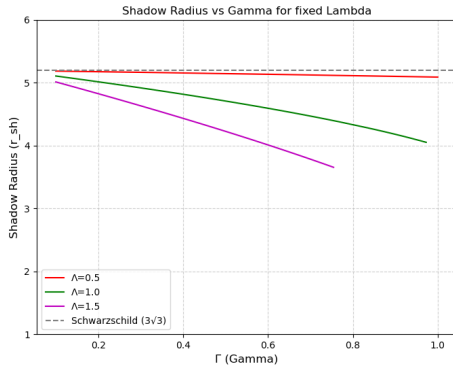
$$\frac{d}{dr} \left(\frac{r^2}{f(r)} \right) = 0 \quad (5.3)$$

For the Schwarzschild case, the photon sphere radius is found to be $3GM$. Photon sphere radius is also calculated for the modified metric and the plots are obtained for studying the variation in these parameters.

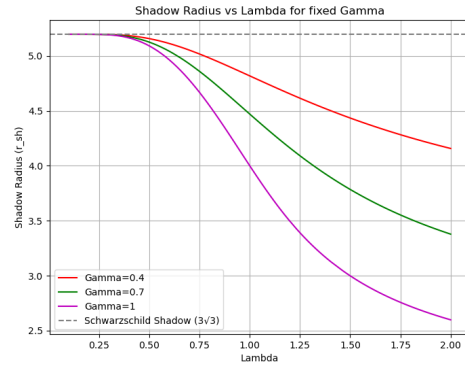
The graphs show how photon sphere radius is dependent on the parameters Γ and λ . R_{ph} tends to decrease with increase in Γ . R_{ph} also tends to decrease faster for higher fixed λ values. The decrease is recorded steepest for $\lambda = 1$ which is following the Reissner-Nordstrom blackhole where increasing charge decreases the photon sphere radius.

R_{ph} also tends to decrease with increase in λ . The rate of decrease increases with higher Γ values. The deviation from the general relativity case is mostly seen for the smaller values of λ and the curve tends to get flat when $\lambda \rightarrow 2$.

Both the graph resorts back to the Schwarzschild case when Γ and λ are taken as 0.



(a) Variation in Gamma



(b) Variation in Lambda

Figure 5.2: Shadow Radius variation for (a) Fixed lambda values (b) Fixed Gamma values

The **shadow radius** of a black hole can be defined as the apparent size of the "shadow" which is cast by the black hole's event horizon due to gravitational lensing of light around it (Perlick and Tsupko). For a non-rotating black hole (Schwarzschild case), the shadow is mostly

circular. The shadow radius R_{sh} is related to the photon sphere radius and is given by :

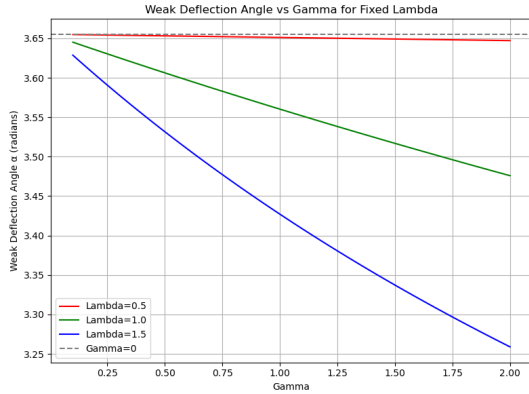
$$R_{sh} = \frac{R_{ph}}{\sqrt{1 - \frac{R_s}{R_{ph}}}} \quad (5.4)$$

In terms of Schwarzschild metric, the shadow radius is given as

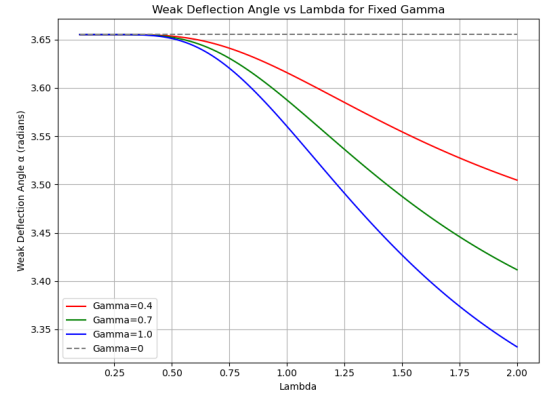
$$R_{sh} = \frac{3\sqrt{3}}{2} r_s$$

Both the graphs show how shadow radius is dependent on Γ and λ values. For fixed values of λ , the shadow radius tends to decrease monotonically with increase in Γ values. The descent is found to be steeper for higher values of λ .

The shadow radius also decreases with an increase in λ values. The decrease is found to be more prominent for smaller values of λ and the deviation minimizes with an increase in λ . It is also observed that the deviation in the shadow radius is more prominent for higher fixed Γ values.



(a) Variation in Gamma



(b) Variation in Lambda

Figure 5.3: Weak Deflection Angle variation for (a) Fixed lambda values (b) Fixed Gamma values

The **Weak Deflection angle** refers to the small angle deviations in the path of light that happen due to light passing through a gravitational lens that is formed by a star or a distant galaxy. For the given metric

$$f(r) = 1 - \frac{2GM}{r} + \frac{\Gamma}{r^{2/\lambda}}$$

In the weak field limit, the Deflection angle can be approximated as

$$\alpha = \frac{2}{b} \int_{r_0}^{\infty} \left[\left(\frac{r}{b} \right)^2 \frac{1}{f(r)} - 1 \right]^{-1/2} \frac{dr}{r} \quad (5.5)$$

where b is the impact parameter and r_0 is the closest approach radius which is related to b by:

$$b = \frac{r_0}{\sqrt{f(r_0)}}$$

For both graphs, the deflection angle is found to decrease with an increase in Γ and λ parameters. The deflection angle decreases with increase in Γ . The decrease is found to be the steepest for the highest fixed value of λ . Γ significantly affects the space-time curvature near the photon path.

The deflection angle is also found to decrease with an increase in λ values. The deviation from the general relativity case is found to be more prominent for smaller λ values. The deviation is also more effective for higher Γ values.

Unlike the Reissner-Nordstorm metric, the additional terms tend to decrease all the lensing parameters such as photon sphere, shadow radius, and Weak deflection angle.

The terms Γ and λ in the modified metric add a positive deformation that falls off with an increase in r , depending on λ . Γ controls the strength of the deformation and λ controls the rate at which the Γ term is decaying with r . The overall term $\frac{\Gamma}{r^{2/\lambda}}$ reduces the curvature of the space-time.

With the increase in Γ and λ , it is observed that all the studied lensing parameters i.e. the photon sphere, shadow radius, and the deflection angle tend to decrease. Γ , and λ tends to influence the space-time curvature, making the gravitational field weak. It can give rise to a modified gravitational scenario where the central object's gravitational influence is effectively diluted. This gravity modification will make it harder to detect objects via lensing as they produce smaller shadows and weak deflections.

CHAPTER 6

GRAVITATIONAL LENSING IN MORRIS-THORNE WORMHOLE METRIC

Considering a general spherically symmetric and static wormhole space-time of Morris Thorne class and is generally written as :

$$ds^2 = -e^{2\phi(r)} dt^2 + (1 - B(r)/r)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (6.1)$$

The Lagrangian $\mathcal{L}$ of a particle is given as :

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \quad (6.2)$$

where λ is an affine parameter.

$$\Rightarrow \mathcal{L} = \frac{1}{2} \left(-e^{2\phi(r)} \left(\frac{dt}{d\lambda} \right)^2 + \left(1 - \frac{B(r)}{r} \right) \left(\frac{dr}{d\lambda} \right)^2 + r^2 \left(\frac{d\theta}{d\lambda} \right)^2 + r^2 \sin^2\theta \left(\frac{d\phi}{d\lambda} \right)^2 \right)$$

For a photon moving in an equatorial plane, i.e. $\theta = \pi/2$,

$$2\mathcal{L} = -e^{2\phi(r)} \left(\frac{dt}{d\lambda} \right)^2 + \left(1 - \frac{B(r)}{r} \right) \left(\frac{dr}{d\lambda} \right)^2 + r^2 \left(\frac{d\phi}{d\lambda} \right)^2$$

As we know, the energy and angular momentum remain conserved, equating the conserved energy and conserved angular momentum, we get

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \dot{t}} &= -e^{2\phi(r)} \dot{t} = -E = \text{constant} \\ \Rightarrow E &= e^{2\phi(r)} \frac{dt}{d\lambda}\end{aligned}\tag{6.3}$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} &= r^2 \dot{\phi} = L = \text{constant} \\ L &= r^2 \frac{d\phi}{d\lambda}\end{aligned}\tag{6.4}$$

Calculating the impact parameter b which is given as $b=L/E$

$$b = \frac{L}{E} = \frac{r^2 d\phi/d\lambda}{e^{2\phi(r)} dt/d\lambda}$$

Substituting the value of $dt/d\lambda = Ee^{-2\phi(r)}$; $d\phi/d\lambda = L/r^2$, we get

$$\begin{aligned}-E^2 e^{2\phi(r)} + \left(\frac{dr}{d\lambda}\right)^2 (1 - B(r)/r)^{-1} + \frac{L^2}{r^2} &= 0 \\ \Rightarrow \left(\frac{dr}{d\lambda}\right)^2 &= (1 - B(r)/r)^{-1} [E^2 e^{2\phi(r)} - L^2/r^2] \\ \left[\cdot \frac{dr}{d\lambda} = \frac{dr}{d\phi} \frac{d\phi}{d\lambda}; \frac{d\phi}{d\lambda} = \frac{L^2}{r^2}\right] \\ \left(\frac{dr}{d\phi}\right)^2 &= \frac{r^4}{L^2} \left(1 - \frac{B(r)}{r}\right) \left[E^2 e^{2\phi(r)} - \frac{L^2}{r^2}\right] \\ \Rightarrow \left(\frac{dr}{d\phi}\right)^2 &= \frac{r^4}{b^2 E^2} \left(1 - \frac{B(r)}{r}\right) E^2 \left[e^{-2\Phi(r)} - \frac{b^2}{r^2}\right] \\ \Rightarrow \left(\frac{dr}{d\phi}\right)^2 &= \frac{r^4}{b^2} \left(1 - \frac{B(r)}{r}\right) \left[\frac{1}{b^2} - \frac{e^{2\Phi(r)}}{r^2}\right] \\ \Rightarrow \frac{dr}{d\phi} &= \pm \frac{r^2}{b} \sqrt{1 - \frac{B(r)}{r}} \sqrt{\frac{1}{b^2} - \frac{e^{2\Phi(r)}}{r^2}}\end{aligned}\tag{6.5}$$

The above equation is the required orbit equation for a photon in the Morris Thorne wormhole metric. To calculate the deflection angle,

$$\Rightarrow d\phi = \pm \frac{b}{r^2} \frac{dr}{\sqrt{1 - \frac{B(r)}{r}}} \left[\frac{1}{b^2} - \frac{e^{2\Phi(r)}}{r^2} \right]^{-1/2}$$

For an inward photon moving towards the turning point r_{tp} , dr is negative. We choose the negative sign to ensure that $d\phi$ is positive. $\therefore$ The inward trajectory is given as :

$$d\phi = -\frac{b}{r^2} \frac{dr}{\sqrt{1 - \frac{B(r)}{r}}} \left[\frac{1}{b^2} - \frac{e^{2\Phi(r)}}{r^2} \right]^{-1/2}$$

The total change in ϕ for photon's path (from infinity to r_{tp}) and back is -

$$\Delta\phi = 2 \int_{r_{tp}}^{\infty} \frac{b}{r^2} \frac{dr}{\sqrt{1 - B(r)/r}} \left[\frac{1}{b^2} - \frac{e^{2\Phi(r)}}{r^2} \right]^{-1/2}$$

The deflection angle α is calculated as

$$\alpha = \Delta\phi - \pi$$

$$\alpha = 2 \int_{r_{tp}}^{\infty} \frac{e^{\phi(r)} dr}{r^2 \sqrt{1 - B(r)/r} \sqrt{\frac{1}{b^2} - \frac{e^{2\Phi(r)}}{r^2}}} - \pi \quad (6.6)$$

where $b=L/E$ is the impact parameter and r_{tp} is the turning point and is related as $b = r_{tp} e^{-\phi(r_{tp})}$. The above equation is the required deflection angle and it diverges logarithmically to infinity as the turning point approaches the photon sphere and results in the formation of relativistic images (Rajibul et al.)

To determine the **photon sphere** for a Morris Thorne wormhole metric, we mainly analyze the geodesic motion of light (null geodesics) in space-time. The Lagrangian is obtained as

$$\mathcal{L} = \frac{1}{2} \left(-e^{2\Phi(r)} \dot{t} + \left(1 - \frac{B(r)}{r} \right) \dot{r} + r^2 \dot{\phi}^2 \right)$$

Substituting the conserved energy and conserved angular momentum into the null condition

$ds^2 = 0$, we get

$$\begin{aligned} \Rightarrow -e^{2\phi(r)}\dot{t}^2 + \left(1 - \frac{b(r)}{r}\right)\dot{r}^2 + r^2\dot{\phi}^2 &= 0 \\ \Rightarrow \dot{r}^2 &= \left(1 - \frac{b(r)}{r}\right) \left(E^2 - \frac{L^2 e^{2\phi(r)}}{r^2}\right) \end{aligned}$$

Defining the impact parameter $b=L/E$, the radial equation becomes -

$$\dot{r}^2 + V_{eff}(r) = 0$$

where the effective potential is given by :

$$V_{eff} = \left(1 - \frac{b(r)}{r}\right) \left(\frac{1}{b^2} - \frac{e^{2\phi(r)}}{r^2}\right)$$

As we know the photon sphere consists of unstable circular orbits. For circular condition, $\dot{r} = 0$ which implies that there is no change in radial motion. For instability condition, $\ddot{r} = 0$ which is the critical point of the effective potential.

From $\ddot{r} = 0$, we have $V_{eff} = 0$ and therefore :

$$\frac{1}{b^2} = \frac{e^{2\phi(r)}}{r^2}$$

For $\ddot{r} = 0$:

$$\begin{aligned} \frac{d}{dr} \left(\frac{e^{2\phi(r)}}{r^2} \right) &= 0 \\ \Rightarrow \frac{d}{dr} \left(\frac{e^{2\phi(r)}}{r^2} \right)_{r=r_{ph}} &= 0 \\ \Rightarrow \phi'(r_{ph}) &= \frac{1}{r_{ph}} \end{aligned} \tag{6.7}$$

Considering a specific redshift function (Tangphati et al.)

$$\phi'(r) = \phi_0 \log \left(1 - \frac{r_0}{r} \right)^\alpha$$

(The redshift function ensures a well-behaved gravitational redshift which is necessary for studying light propagation in wormhole gravitational field)

$$\Rightarrow \alpha \phi_0 \frac{r_0}{r(r - r_0)} = \frac{1}{r_{ph}}$$

$$\Rightarrow r_{ph} = r_0(1 + \alpha\phi_0) \quad (6.8)$$

Numerical solutions show that for static wormholes, the photon sphere ranges from $2.2M \leq r_{ph} \leq 3.5M$.

The **critical impact parameter** is defined as the photon's angular momentum (L) to its energy (E) and is given by $b=L/E$.

$$\begin{aligned} \frac{1}{\bar{b}} &= \frac{e^{2\phi(r)}}{r_{ph}^2} \\ \Rightarrow \bar{b}_{ph} &= \frac{r_{ph}}{e^{\phi(r_{ph})}} \end{aligned} \quad (6.9)$$

Substituting the value of r_{ph} , we get

$$\begin{aligned} \bar{b}_{ph} &= \frac{r_0(1 + \alpha\phi_0)}{\left(1 - \frac{r_0}{(1+\alpha\phi_0)}\right)^{\alpha\phi_0}} \\ \bar{b}_{ph} &= \frac{r_0(1 + \alpha\phi_0)}{\left(\frac{\alpha\phi_0}{1+\alpha\phi_0}\right)^{\alpha\phi_0}} \end{aligned} \quad (6.10)$$

The above equation is the shadow boundary of the wormhole which is seen by an observer.

Considering two sets of different parameters (Takol et al.).

Case 1 : $b_1(r)$, parameter $\alpha = 1; \beta = 1; \phi_0 = 1; r_0 = 1$

Case 2 : $b_2(r)$, parameter $\alpha = 1.5; \gamma = 1; \phi_0 = 1; r_0 = 1$

Thus the shadow radius ranges from $4.0m \leq R_{shadow} \leq 5.3M$

CHAPTER 7

COMPARATIVE ANALYSIS OF LENSING SIGNATURES

There exist several factors that distinguish a wormhole from a black hole. Let us consider a static non-rotating spherically symmetric wormhole having the redshift function $\phi'(r) = \phi_0 \log \left(1 - \frac{r_0}{r}\right)^\alpha$ to model the wormhole geometry.

The photon sphere radius depends mainly on space-time geometry. For a Schwarzschild black hole, the photon sphere is at a fixed radius

$$r_{ph} \approx \frac{3GM}{c^2} \approx 3M$$

The photon sphere radius for a static traversable wormhole (Morris Thorne) is

$$r_{ph}^2 = e^{2\phi(r_{ph})} \bar{b}^2$$

The throat of the wormhole does not appear as a maximum of V_{eff} , since $dV_{eff}/dr \neq 0$. In proper radial coordinates,

$$l(r) = \pm \int_{r_0}^r \frac{dr}{r_0 \sqrt{1 - b(r)/r}}$$

the effective potential has a maximum at the throat, satisfying :

$$V_{eff} = E^2; \frac{dV_{eff}}{dl}|_{r_0} = 0; \frac{d^2V_{eff}}{dl^2}|_{r_0} < 0$$

Unlike black holes, the photon sphere radius depends on the specific redshift and shape functions. The absence of an event horizon in the wormhole space-time can lead to a possibility that some

light rays can pass through the wormhole throat which can potentially create additional photon orbits or modify the stability of the photon sphere.

blackholes have a single well-defined photon sphere (or a range of photon spheres for Kerr) which are terminated by the event horizon where light is completely absorbed. Wormhole lacks an event horizon which allows the light to traverse the throat and create multiple photon spheres which can lead to complex lensing patterns.

For the shadow radius, (R_{shadow}) is the apparent size of the dark region that is cast by the object as seen by a distant observer. For a Schwarzschild-like black hole, the shadow radius is found to be

$$R_{shadow} \approx 3\sqrt{3}M \approx 5.2M$$

The shadow cast by a Schwarzschild black hole is circular due to spherical symmetry with light rays crossing the photon sphere falling in the event horizon which creates a sharp, dark boundary.

The shadow radius for a traversable wormhole is found by the following expression :

$$R_{shadow} = \bar{b}_{ph} = \frac{r_{ph}}{e^{\phi(r_{ph})}}$$

with numerical results ranging from $4.0M \leq R_{shadow} \leq 5.3M$. The shadow is found to be symmetric but its size varies with the redshift and shape function. Unlike black holes, the absence of an event horizon allows light rays to pass through the throat which potentially reduces the shadow size and darkness, creating secondary images, and leading to deviations from a perfect circle.

Black hole shadows are sharply defined by the event horizon, producing a dark, nearly circular (Schwarzschild) or asymmetric (Kerr) region with a single photon ring. Wormhole shadows are less dark and may exhibit complex structures due to light passing through the throat, leading to additional lensing effects like multiple photon rings. These features may arise as deviations in the shadow size, shape, and intensity distribution.

The deflection angle for a Morris Thorne wormhole is found to be :

$$\alpha = 2 \int_{r_{tp}}^{\infty} \frac{e^{\phi(r)} dr}{r^2 \sqrt{1 - B(r)/r} \sqrt{\frac{1}{b^2} - \frac{e^{2\phi(r)}}{r^2}}} - \pi$$

Where $b=L/E$ is the impact parameter and r_{tp} is the turning point and is related as $b = r_{tp}e^{-\phi r_{tp}}$. From the paper (Shaikh et al., “A novel gravitational lensing feature by wormholes”), it is found that the deflection angle diverges logarithmically to infinity as the turning point approaches a photon sphere. For a Schwarzschild black hole, the deflection angle is found to be

$$\alpha = \frac{4GM}{b}$$

For $b < 3\sqrt{3}M$, light rays are captured by the event horizon at $r=2GM$ and no deflection angle is observed as light does not reach the observer. For the case of wormholes ($b < b_{ph}$), the deflection angle decreases after diverging at $b = b_{ph}$ but diverges again as $b \rightarrow b_{ph}$ due to effective photon sphere at the throat. The lack of an event horizon in the wormhole allows the light to pass through the throat orbit at r_o , potentially escaping the observer.

For the Schwarzschild case, the deflection angle is found solely dependent on M with a fixed photon sphere and a critical impact parameter. But this differs from the case of the wormhole as the deflection angle depends on the redshift function $\phi(r)$ and shape function $b(r)$ which introduces variability. Both functions affect the shape and existence of the throat's effective photon sphere which leads to different deflection profiles. All these unique features which are influenced by the wormhole's space-time geometry and the throat's role as an effective photon sphere offer potential observational signatures for distinguishing wormholes from black holes.

CHAPTER 8

RESULTS AND DISCUSSION

This study examines gravitational lensing in Schwarzschild black holes, Kalb-Ramond modified black holes, and Morris-Thorne wormhole space-times, revealing unique features that distinguish these objects through their photon sphere radius, shadow radius, and weak deflection angle.

In the Schwarzschild space-time, the black hole's photon sphere, shadow radius, and lensing effects are well-defined by general relativity. The photon sphere, where light follows unstable circular paths, is fixed at approximately $3M$ (where M is the mass in geometric units). The shadow radius, which shows the dark region seen by a distant observer, is sharp and circular at about $5.2M$, caused by the event horizon at $2M$ that traps light. The deflection angle, which measures how much light bends, is $4GM/b$ in the weak field limit, but no deflection occurs for impact parameters b less than $3\sqrt{3} M$, as light is captured by the event horizon.

In Kalb Ramond gravity, the Schwarzschild metric includes a modification term controlled by the parameters Γ (strength) and λ (decay rate). The overall term $\Gamma/r^{2/\lambda}$ modifies the gravitational potential which affects the space-time geometry and the behavior of light. The correction term strengthens the potential at small r , pushing the photon sphere orbit outward. This outward shift increases the critical impact parameter for photon orbits which results in the observation of a larger shadow radius. The shadow radius is determined by the photon sphere and the correction term increases the deviation in the r_{sh} as compared to the Schwarzschild case, indicating that the modified gravity enhances the bending of light near the black hole. For the weak deflection angle, the correction term modifies the gravitational potential at large r , which increases the deflection as compared to the Schwarzschild case where $\alpha = 4GM/b$ in the weak field limit. The larger values of α show significant deviation from the general relativity case which potentially affects

the phenomena like gravitational lensing. Γ determines the magnitude of the correction to the Schwarzschild potential and acts as a coupling constant. Γ is generally responsible for enhancing the deviation in the observables from the general relativity and is dependent on λ . λ determines the decay of the correction term via the exponent $2/\lambda$ and controls the range of modification with smaller λ making the correction dominant near the black hole which significantly affects the near horizon phenomena such as photon sphere and shadow radius. Γ and λ together can be used to test the deviations from the general relativity where smaller λ and larger Γ values are found to produce the most significant deviations. These deviations have significant observational implications which are potentially detectable through the black hole shadow and gravitational lensing effects.

For the Morris-Thorne wormhole, the photon sphere radius depends on the redshift function $\phi(r) = \phi_0 \log(1 - r_0/r)^\alpha$ and shape function $b(r)$, ranging from 2.2 M to 3.5 M. Wormholes can have multiple photon spheres, including one at the throat. The shadow radius, given by $r_{ph}/e^\phi(r_{ph})$, ranges from 4.0M to 5.3M and is less dark because light can pass through the throat, creating additional images. The deflection angle becomes very large near the photon spheres or throat, leading to unique patterns like two sets of Einstein rings: one loosely spaced from the outer sphere and one tightly packed at the throat. The absence of an event horizon in the Morris Thorne wormhole contrasts with the Schwarzschild black hole feature and can be used to test general relativity. Wormhole-like shadows can indicate scenarios like modified gravity or geometries beyond general relativity, especially in strong gravitational fields near the throat. Unique lensing features such as multiple photon rings or relativistic images can be detected by a high-resolution instrument such as the Event Horizon Telescope[EHT].

In comparison, Schwarzschild black holes have fixed, clear lensing features, Kalb-Ramond black holes show weaker effects due to reduced curvature, and wormholes display varied, complex patterns with multiple photon spheres and fainter shadows. These differences demonstrate that gravitational lensing can effectively distinguish between black holes and wormholes.

CHAPTER 9

CONCLUSION

The project explored the optical properties of a black hole in general relativity, modified gravity, and for a traversable wormhole and conducted a comparative analysis with the lensing feature of a Schwarzschild black hole. Orbit equation, photon sphere radius and deflection angle in the weak field limit are derived in order to throw light on the similarities and distinct observational signatures between black holes and wormholes. The comparative analysis highlights that while Schwarzschild black holes produce well-defined lensing signatures, Kalb Ramond exhibits weaker lensing effects due to modified curvature, and Morris Thorne wormhole displays varied and complex lensing patterns. The difference in their optical properties provides a potential observational field to distinguish a wormhole from a black hole. Further observation with advanced imaging instruments like Event Horizon Telescope (EHT) could detect these distinctive lensing signatures such as multiple photon rings or relativistic images. Optical properties and lensing signatures are some of the fundamental observational tools in physics and can be used in exploring complex space-time geometries.

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